SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING - CSA17 ARTIFICIAL INTELLIGENCE LAB

Day 1 - Program 7	Course Outcome: CO3	Language: Python 3

Question

Write the python program to implement BFS.

Aim

To implement the Breadth First Search graph traversal.

Algorithm / Explanation

BFS uses a FIFO queue. The start node is enqueued; repeatedly the front node is removed, visited and all its unvisited neighbours are enqueued.

Nodes are therefore visited level by level, which also gives the shortest path (fewest edges) in an unweighted graph. Time complexity O(V + E).

Program

```
# Breadth First Search (BFS)
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['F'],
    'F': []
}

def bfs(graph, start):
    visited = [start]
    queue = deque([start])
    while queue:
        node = queue.popleft()
        print(node, end=" ")
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.append(neighbour)
                queue.append(neighbour)

print("BFS traversal starting from A:")
bfs(graph, 'A')
print()

def bfs_path(graph, start, goal):
    queue = deque([[start]])
    visited = {start}
    while queue:
        path = queue.popleft()
        if path[-1] == goal:
            return path
        for n in graph[path[-1]]:
            if n not in visited:
                visited.add(n)
                queue.append(path + [n])

print("Shortest path A -> F:", bfs_path(graph, 'A', 'F'))
```

Output

```
BFS traversal starting from A:
A B C D E F
```

```
Shortest path A -> F: ['A', 'C', 'F']
```

Result

Thus the Python program was executed successfully and the output was verified.